Siddharth Kapoor

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siddkapo.github.io

Education

2025–Present **PhD Computer Science**,
The University of Utah, CGPA – 4/4

2018–2022 **BE Computer Science**,
Birla Institute of Technology & Science, Pilani, CGPA – 8.16/10

Manuscripts Under Submission

1. **Dishonest-Majority Fair MPC from Well Studied Assumptions**,
S. Kapoor, P. Soni
2. **Making Maliciously Secure Graph Computation Practical**,
S. Kapoor, S. Patil, A. Patra, B. R. Gopal
3. **GraSP: Secure Collaborative Graph Processing Made Scalable**,
S. Kapoor, N. Koti, V. B. Kukkala, A. Patra, B. R. Gopal, https://eprint.iacr.org/2025/590
4. **SMILIES: Secure eMaIL InfrastructurE and Services**,
S. Kapoor, S. Murthy, S. Murthy, A. Patra, B. R. Gopal, R. Ramesh

Research Experience

Jul'24–Jul'25 **Research Associate**,
Indian Institute of Science (IISc), Bengaluru

- Worked as an RA in the CrIS Lab under Prof. Arpita Patra, in the Department of CSA at IISc Bengaluru.
- Implemented Secure Multi-Party Computation (MPC) based protocols for problems related to Privacy Preserving Machine Learning and Secure Graph Computation.

Industry Experience

Jul'22–Jul'24 **Software Development Engineer**,
Amazon, Bengaluru

- Implemented handwriting recognition and gesture support for the stylus on Amazon Fire tablets.
- Worked on enabling app compatibility for VegaOS based Amazon Fire TVs.
- Fixed critical bugs in the Android and Linux application frameworks.

Jan'22–Jun'22 **Software Development Engineer Intern**,
Amazon, Chennai

- Implemented predictive touch algorithms to reduce perceived touch latency while using a stylus.
- Developed a novel metric to measure reactivity of predicted outputs from different predictive touch algorithms.

May'21–Jul'21 **Technical Undergraduate Intern**,
Cisco, Bengaluru

- Improved logging for the Cisco Identity Services Engine for faster troubleshooting.

Selected Projects

1. **Multivariate Polynomial Encryption Schemes**,
   Design Project, Prof. Abhishek Mishra, BITS Pilani
   - Implemented the paper titled 'Multivariate Encryption Schemes Based on Polynomial Equations over Real Numbers' from scratch in C++.
   - Dived deep into the underlying mathematics of multivariate cryptography to develop a complete understanding of the project.
2. **Privacy Preserving Machine Learning**,
   Study Project, Prof. Ashutosh Bhatia, BITS Pilani
   - Studied different privacy-preserving techniques for machine learning models, with a special focus on homomorphic encryption.

Technical Skills

Languages C/C++, Go, Java, Python, SQL
Technologies Linux, Android, AWS, Git, OpenMP

Relevant Coursework

Cryptography	Algorithms	Theory of Computation
Discrete Math	Graph Theory	Probability & Statistics
Quantum Information	Machine Learning	Distributed Systems

Awards and Achievements

2017 **Kishore Vaigyanik Protsahan Yojana (KVPY) Fellowship**,
Indian Institute of Science, Bengaluru
Qualified for the KVPY fellowship in the SX stream (Declined).

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